OSELTAMIVIR

Start within 48 h

Pregnancy: recommended

CHEMICAL NAME Ethyl (3R,4R,5S)-4-acetamido-5-amino-3-(pentan-3-yloxy)cyclohex-1-ene-1-carboxylate (phosphate)

CLASS Neuraminidase inhibitor antiviral (influenza)

TRADE NAMES Tamiflu · Oseltamivir phosphate

THERAPEUTIC USES / INDICATIONS

- Treatment of influenza A and B — best within 48 h of symptoms; any time in severe/hospitalised or high-risk patients.
- Post-exposure and seasonal prophylaxis in high-risk groups.
- Avian influenza (H5N1).

ADMINISTRATION

- **PO** capsules or suspension, with food (reduces nausea).
- Capsules can be opened and mixed with sweet liquid.

DOSAGES

- **Treatment (adult & child >40 kg):** 75 mg twice daily for 5 days.
- **Children ≥1 yr:** ≤15 kg 30 mg; 15–23 kg 45 mg; 23–40 kg 60 mg — twice daily.
- **Infants <1 yr:** 3 mg/kg twice daily.
- **Prophylaxis:** same dose once daily for 10 days (post-exposure).
- CrCl 30–60: 30 mg twice daily; 10–30: 30 mg once daily.

PHARMACOKINETICS

- **Peak:** 3–4 h (active metabolite). Bioavailability ~80% as active oseltamivir carboxylate.
- Prodrug converted by liver esterases.
- Excreted in urine. **t½:** 6–10 h.

PHARMACODYNAMICS

- Inhibits viral **neuraminidase** → newly formed virus particles cannot be released from infected cells → limits spread.
- Shortens illness by ~1 day; reduces complications in high-risk groups.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting (reduced with food).
- Headache, abdominal pain.
- Diarrhoea.

ADVERSE EFFECTS (SERIOUS)

- Neuropsychiatric events (delirium, abnormal behaviour, self-injury); mainly children/adolescents (causality uncertain).
- Severe skin reactions (rare).
- Hepatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** severe renal impairment.

KEY INTERACTIONS

- **Live attenuated influenza vaccine (nasal):** do not give within 48 h before to 2 weeks after oseltamivir.
- Inactivated vaccine; no interaction.

NURSING MANAGEMENT

- Start promptly; do not wait for test result in high-risk patients.
- Droplet precautions; isolate where possible.
- Monitor respiratory status, temperature, hydration.
- Observe children for unusual behaviour.
- Check renal function for dosing.

PATIENT EDUCATION

- Take with food; complete 5 days.
- Rest, drink fluids, stay home until 24 h fever-free.
- Report confusion or unusual behaviour (children).
- Annual flu vaccination is still important.

KEY POINTS

- Neuraminidase inhibitor for influenza A & B.
- Nausea — take with food.
- Best within 48 h; 75 mg BD × 5 days.
- Safe and recommended in pregnancy.

SOURCES CHECKED **[247]** Oseltamivir 30 mg Hard Capsules (UK SmPC) **[248]** Oseltamivir Phosphate, oral (US label) **[249]** Oseltamivir, oral (US label)

Full references p. 432–457